

Effectiveness and tolerability of a standardized extract from Hibiscus sabdariffa in patients with mild to moderate hypertension: a controlled and ...

In order to compare the antihypertensive effectiveness and tolerability of a standardized extract from *Hibiscus sabdariffa* with captopril, a controlled and randomized clinical trial was done. Patients from 30 to 80 years old with diagnosed hypertension and without antihypertensive treatment for at least 1 month before were included. The experimental procedure consisted of the administration of an infusion prepared with 10 g of dry calyx from *H. sabdariffa* on 0.51 water (9.6 mg anthocyanins content), daily before breakfast, or captopril 25 mg twice a day, for 4 weeks. The outcome variables were tolerability, therapeutic effectiveness (diastolic reduction $>$ or $=$ 10 mm Hg) and, in the experimental group, urinary electrolytes modification. Ninety subjects were included, 15 withdrew from the study due to non-medical reasons; so, the analysis included 39 and 36 patients from the experimental and control group, respectively. The results showed that *H. sabdariffa* was able to decrease the systolic blood pressure (BP) from 139.05 to 123.73mm Hg (ANOVA $p < 0.03$) and the diastolic BP from 90.81 to 79.52mm Hg (ANOVA $p < 0.06$). At the end of the study, there were no significant differences between the BP detected in both treatment groups (ANOVA $p > 0.25$). The rates of therapeutic effectiveness were 0.7895 and 0.8438 with *H. sabdariffa* and captopril, respectively (χ^2 , $p > 0.560$), whilst the tolerability was 100% for both treatments. A natriuretic effect was observed with the experimental treatment. The obtained data confirm that the *H. sabdariffa* extract, standardized on 9.6mg of total anthocyanins, and captopril 50 mg/day, did not show significant differences relative to hypotensive effect, antihypertensive effectiveness, and tolerability.